

# ARYAN SRIVASTAVA

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## PROFILE

Second-year Computer Science student focused on CUDA-adjacent GPU systems, GPU dataframe internals, LLM inference benchmarking, backend infrastructure, and production-grade open-source development.

Contributes to large C++/Python and Rust codebases, writes tests, fixes behavioral inconsistencies, and works across backend, AI, and HPC systems.

## EDUCATION

Manipal Institute of Technology, Bengaluru

B.Tech in Computer Science and Engineering | Expected January 2028

## OPEN SOURCE CONTRIBUTIONS

NVIDIA RAPIDS - cuDF

- PR #20747: Boolean Casting Consistency Fix, merged.
- Fixed behavioral inconsistency between cuDF and Pandas; modified Python/C++ interop logic and added regression tests.
- PR #20862: Hybrid Scan API, merged.
- Added a hybrid scan API in libcudf C++ for all-true row masks with memory-safe null handling and unit tests.

Polars - High-Performance DataFrame Library (Rust)

- PR #27669: FixedRingBuffer Allocation Provenance, merged.
- Fixed a Rust allocation-provenance soundness issue in polars-utils by preserving the original Vec allocation.

## PROJECTS

MAHE Mobility Challenge 2026 - 1st Place, AI Track | Backend Architect

- Built a connected-vehicle alert pipeline with local Phi-3 classification, Node.js/WebSocket triage, a heatmap, and recovery summaries.

LLM Inference Benchmarking | Sarvam 30B FP8

- Repo: [github.com/aryansri05/indicservebench](https://github.com/aryansri05/indicservebench)
- Benchmarked Sarvam 30B FP8 on H100 SXM using 240 measured inference requests across English, Hindi, Tamil, and Hinglish/code-mixed prompts.
- Tracked tokenization behavior, mean latency, P90/P95 latency, and throughput to analyze serving performance and deployment tradeoffs.
- Additional T4 and Apple M2 baselines are being expanded.

Paper Trading Platform

- Full-stack options simulator with React, Node.js/Express, PostgreSQL, JWT auth, portfolio tracking, and real-time market data.

## TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, Bash

AI / ML: LangGraph, FastMCP, ChromaDB, Ollama, Llama 3, RAG, LLM agents

HPC / GPU: CUDA, cuDF, RAPIDS AI, LLM inference benchmarking, Apple Silicon unified memory, SLURM

Frameworks: React.js, Node.js, Express, Pandas, NumPy, Pytest

Tools: Git, GitHub Actions, Docker, CI/CD, PostgreSQL, GraphQL, VS Code